

RayCastED: A RayCast End-to-end Detection Model for Lightweight and Boundary-Aware Nuclei Detection

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Clinical Background: TILs Assessment

- **Tumor-infiltrating lymphocytes (TILs)** are established prognostic biomarkers in multiple cancers
- International TILs Working Group guidelines recommend **area-based scoring**: percentage of stromal area occupied by lymphocytes
- **Critical requirement**: accurate boundary delineation, not just centroid detection

The Challenge

Manual assessment is:

- Time-consuming
- Subject to inter-observer variability
- Impractical for whole-slide images

$$\text{sTILs (\%)} = \frac{\text{Area of mononuclear inflammatory cells in stroma}}{\text{Total intratumoral stromal area}} \times 100$$

Clinical Background: Immune Cell Density Scoring

- **Immune Cell Density** quantifies the absolute number of immune cells within a defined tissue region.
- Represents a shift from traditional visual area estimation to precise **single-cell enumeration**.
- **Critical requirement:** robust **instance segmentation** to accurately detect, separate, and classify individual nuclei.

The Challenge

Manual single-cell counting is:

- Practically impossible at scale
- Highly prone to human error
- Requires digital AI pipelines for whole-slide analysis

$$\text{Immune Cell Density} = \frac{\text{Total count of detected immune cells}}{\text{Analyzed tissue area (mm}^2\text{)}}$$

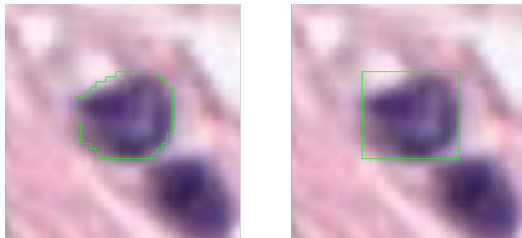
The Clumped Cell Problem

- Densely clustered nuclei merge in segmentation masks
- **Result:** Cell undercounting
- Existing solutions require costly post-processing
- Non-Max Suppression (NMS) suppresses overlapping predictions (risky in clumped regions)

Key Issues

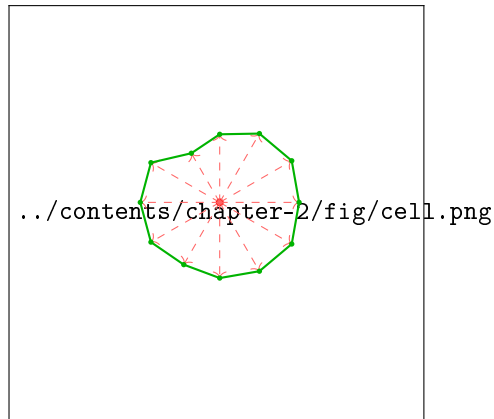
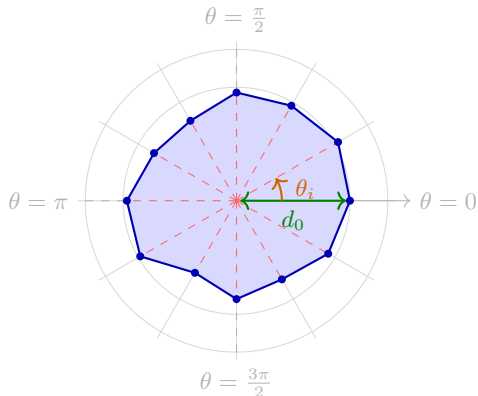
- 1 Instance separation overhead
- 2 NMS-based suppression errors
- 3 Deployment inaccessibility (transformers)

The Accuracy-Deployment Trade-off



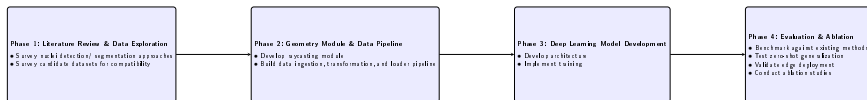
Approach	Pros	Cons
Instance Segmentation (U-Net, Mask R-CNN)	Accurate boundaries	Computationally expensive (unsuitable for edge)
Transformer-based (CellViT, DINO)	Strong performance Good separation	Quadratic complexity (memory/compute intensive)
Object Detection (YOLO, Faster R-CNN)	Fast, lightweight Edge-deployable	No boundary geometry (insufficient for TILs)

Star-Convex Polygons for Cell Representation



- **Left:** Polar coordinate representation with centroid $\mathbf{c}$ and radial distances d_i
- **Right:** Applied to real cell with 12 rays at fixed angles θ_i
- Preserves boundary geometry while maintaining per-cell separation

Research Workflow



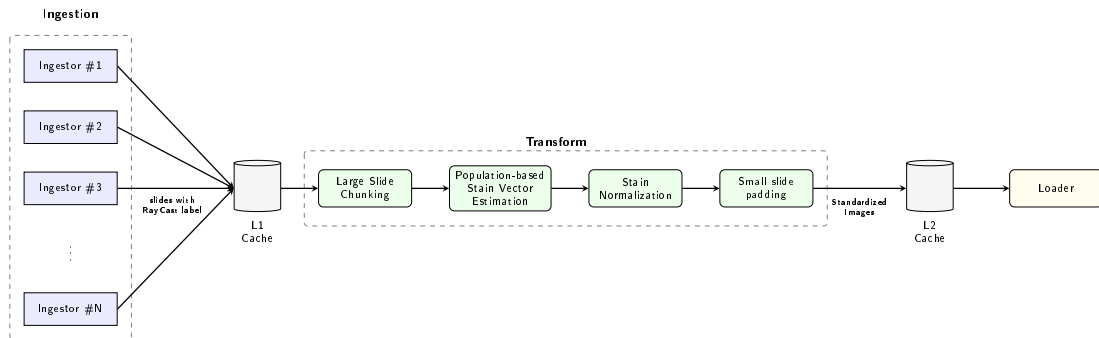
Datasets

Dataset	Purpose	Tissue	Format
PanNuke	Primary benchmarking	Multi-organ	Pickled bytes
PUMA	TILs detection	Melanoma	TIFF, GeoJSON
panopTILs	TILs detection	Breast	PNG
MoNuSAC	TILs detection	Multi-organ	Pickled bytes

Table: Summary of datasets used in this study.

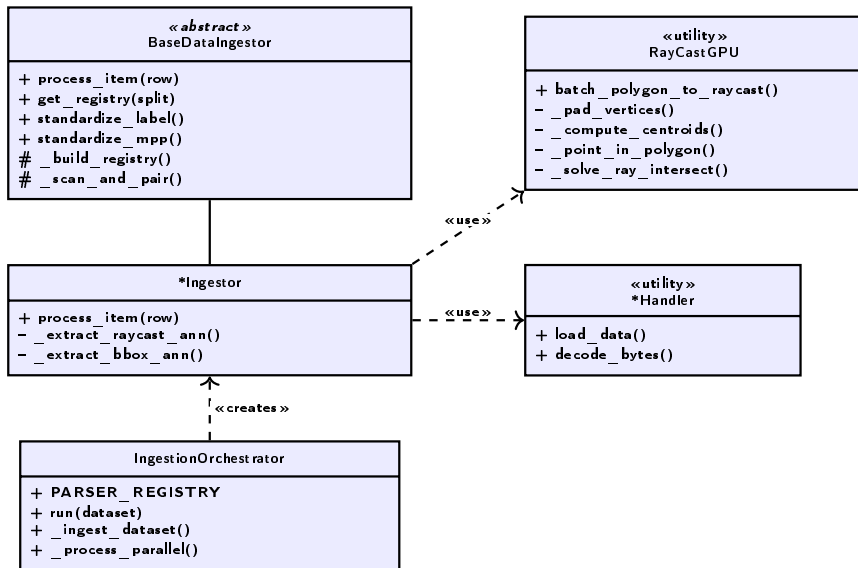
- H&E stained histopathology images
- ~ 0.25 MPP resolution
- Multiple tissue types for generalization

Data Preprocessing Pipeline

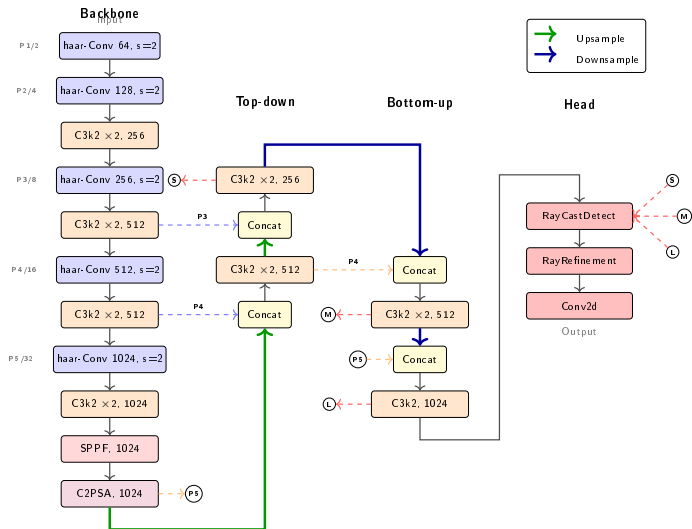


- Multi-format dataset handling (TIFF, PNG, pickled bytes)
- Patch extraction and augmentation
- Star-convex annotation generation

Data Ingestion Design



RayCastED Architecture



Key Innovations

1. RayCast Representation

- Centroid $(x, y) + N$ radial distances
- Lightweight, boundary-aware representation

2. End-to-End Assignment

- Learned bipartite matching
- Eliminates NMS dependency

3. Lightweight FCN Backbone

- Efficient wavelet-enhanced feature extraction
- TensorRT/ONNX compatible

Evaluation Procedure

